

Eli M. Zerr

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Education

Stanford University

Intended B.S. in Mathematics, B.A. in Political Science

Stanford, CA

September 2025 – Present

- Coursework: *Computer Organization and Systems, Applied Matrix Theory, Linear Algebra and Matrix Theory, Fundamental Concepts of Analysis, Introduction to Probability Theory, Programming Abstractions, Multivariable Calculus, Is Privacy Dead?: Privacy, Surveillance, and Freedom in the Digital Age, Ethical Theory, Liberalism and its Critics, Liberalism and Populism in the American Political Economy*
- Anticipated coursework (winter/spring 2026-2027): *Introduction to Optimization, Machine Learning, Graph Theory, Economic Analysis I, Digital Systems Architecture, Introduction to Economics*
- 3.789 cumulative GPA

University of North Dakota

Non-Degree-Seeking Student

Grand Forks, ND

August 2024 – June 2025

Experience

Independent Aviation Analyst

January 2026 – August 2026

- Initiated discussion with the mayor of Grand Forks, ND, to provide additional citizen perspectives on local initiatives to attract additional flights and airlines to Grand Forks International Airport (GFK)
- Analyzed USDOT data (DB1B/C, T-100), publicly available schedule data, and airport authority market study information to identify passenger leakage, travel patterns, connectivity gaps, and potential air service growth opportunities
- Delivered a 50-page report upon the mayor's request, translating findings into potential strategies for local officials to attract and retain new air service
- Followed up in August 2026 with new information following material shifts in observed passenger demand that confirm key report findings

Founder

Septerrex Technologies, LLC

April 2024 – Present

Stanford, CA/Grand Forks, ND

- Collaborated with instructors, administrators, and students to understand diverse needs and bring a novel data-driven approach to music education
- Developed *lessonscheduler.ai*, a scheduling and data collection web app serving nearly 500 music students, teachers, and public school administrators
- Worked with administrators to leverage data and optimize program staffing by 8-10%
- Managed regulatory compliance (FERPA) and integration with existing IT

Research Intern (CURIS Program)

Department of Computer Science, Stanford University

June 2026 – August 2026

Stanford, CA

- Collaborated with faculty, stakeholders, and other interns to develop InLecture, an interactive AI platform for use in higher education, in 10 weeks
- Served as technical lead for the development process, reviewing and testing both human and AI-generated code
- Built internal data classification methods and implemented automated data analysis and visualization to generate instructor and student insights
- Deployed app for fall 2026 pilot program in introductory Stanford computer science courses, serving approximately 250 students

Research Affiliate

Oncel Research Group, University of North Dakota

June 2023 – August 2024

Grand Forks, ND

- Condensed-matter physics lab work
- Designed custom 3D-printed components and electronics to test feasibility of low-cost STM construction

Skills

Communication: Microsoft Office (Word, PowerPoint)

Data and Analytics: Excel, Python (Pandas, NumPy, Matplotlib, Seaborn), SQL/Relational Databases (SQL Server, PostgreSQL, SQLite), Excel

Software and Systems: Flask, SQLAlchemy, C++, C, JavaScript, HTML

Design & Manufacturing: Computer-Aided Design (Fusion 360, OnShape), PCB Design (KiCad), Additive Manufacturing

Other: Project Management, Systems Modeling

Leadership & Policy Engagement

NDDPI Student Cabinet Member

North Dakota Department of Public Instruction

January 2021 – August 2025

Bismarck, ND

- Selected for three consecutive terms to advise the state superintendent on issues relevant to students
- Conducted broad student outreach in order to craft actionable state-level policy recommendations
- Chaired a subcommittee aiming to address root causes of the growing student mental health crisis

Strategic Planning Committee Member

Grand Forks Public Schools

August 2024 – November 2024

Grand Forks, ND

- Invited by Superintendent of Schools Dr. Terry Brenner to advise on district-wide strategy

Service

IntroSems Advisory Committee Member

Office of the Vice Provost for Undergraduate Education

April 2026 – Present

Stanford University

- Selected to serve on an advisory committee for introductory seminars (IntroSems)

Selected Papers and Presentations

Defining “Essential”: A Framework and Metrics for Evaluating Rural Air Service Subsidies

Eli Zerr

- June 2026 term paper for Numbers and Metrics: Rhetoric of Calculation and Quantification, Stanford University
- Nominated for the Boothe Prize for first-year writing

InLecture: An End-to-End AI-Powered Interactive Lecture Platform

Eli Zerr, Joowan Park, Chris Gregg, and Nick Troccoli; Department of Computer Science, Stanford University

- August 2026 Stanford CURIS Program Poster Presentation

Innovation and Collaboration in Support of Music Education

Eli Zerr, Septerrex Technologies, LLC

- March 2026, Invited Presenter, Annual NDMEA Conference

Personal Projects

USDOT Aviation Data Pipeline

July 2026 – Present

- Developing a unified ingestion and processing pipeline for current and historical USDOT aviation datasets to support continued personal work
- Reconciling schema differences across otherwise fragmented datasets in a unified SQL database and integrating aviation data with road network data to enable multimodal analysis of passenger behavior

Multi-Node Geographically Distributed Network Storage System

June 2026 – Present

- Built out over summer 2026, involves multiple Raspberry Pi-powered network storage servers linked together across the open internet with WireGuard tunnels
- Custom configuration of various software services creates seamless storage synchronization between nodes, allowing for defined redundancy levels on a per-dataset basis
- Custom enclosure design for each unit involving 3D-printing, lasercutting, etc.

6502neo Development System

February 2023 – June 2023

- Designed a custom 8-bit SBC based on the 65c02 processor to deepen computer hardware and electronics design knowledge
- Fully custom PCB design, system memory layout, and interface system
- Custom code in 6502 assembly to implement basic functions for a workbench development platform